



Systematic Review of AI-Driven Personalization in Serious Games for Teaching at the Right Level in Morocco [†]

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Abstract: Digitalization has shaped every part of our world, from education to healthcare, from finance to entertainment, and from manufacturing to social interactions. This digital transformation has put tremendous power in the hands of individuals and delivered even better results than sticking to old, traditional ways. Such effective digitalization cannot be achieved easily; a good digitized system is the product of a lot of effort. In this paper, we explore the state-of-the-art advancements in smart education technologies, analyze existing digital solutions, and outline considerations for developing a tailored system to digitalize the Teaching at the Right Level (TaRL) assessment for Moroccan primary education. Our approach aims to bridge skill gaps by integrating serious games, adaptive learning techniques, and real-time analytics to enhance assessment effectiveness and ease educators' workloads.

Keywords: serious games; digital learning; adaptive learning; gamification; learning analytics; education technology; game-based learning; real-time assessment; TaRL



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1. Introduction

Students often face a serious challenge known as academic mismatch, which refers to the gap between a student's actual learning level and their assigned school grade. This problem is widespread globally and has been particularly prevalent in Morocco, where a significant proportion of learners advance through school without mastering foundational skills. According to the World Bank, in the post-COVID-19 era [1], more than 66% of Moroccan 10-year-olds struggle with basic reading comprehension, pointing to a deeper issue in foundational education. Many factors contribute to this issue, including unfinished learning, gaps they may not even be aware of, and emotional or social challenges that cause them to fall behind or feel too shy to seek help. Neglecting this issue and failing to address it early on can lead to even greater problems, such as disengagement, lack of confidence, and, ultimately, high dropout rates [2].

To tackle this issue, Morocco has recently adopted the Teaching at the Right Level (TaRL) approach in primary schools, a method initially developed by Pratham in India [3]. This strategy directly addresses learning gaps by first assessing students' actual abilities rather than assuming alignment with their school grade. It then delivers several weeks of targeted instruction by grouping learners according to their real competencies and areas needing improvement. A recent systematic review highlighted the global effectiveness

of TaRL in improving literacy and numeracy across various contexts and emphasized its adaptability to low-resource settings [4].

The TaRL approach has also been used in Southeast Asia, where it was successfully implemented into the All SMART Children (SAC) model in Indonesia, leading to a 35% improvement in literacy after only two weeks of targeted instruction [5].

Applying TaRL in Moroccan primary schools has helped to narrow learning gaps and allowed students to catch up with their peers. However, although the method has gradually contributed to improvements in the education system, educators, especially teachers, have struggled to maximize its impact while managing their regular responsibilities alongside the additional workload that comes with the TaRL approach. The adjustment process remains demanding due to the method's novelty and the lack of sufficient support materials or well-defined implementation plans.

While TaRL has shown promising results in various global contexts, its digitized implementation, especially in the context of nations with bilingual, infrastructure-constrained systems like Morocco, remains underexplored. This study presents a digitized version of TaRL tailored to Moroccan primary education. The goal is to support educators, simplify instructional delivery, and improve learning outcomes through the strategic use of educational technologies. It also examines relevant digital interventions, particularly those that use personalized educational games, to identify features that can strengthen the rollout of a national digitized TaRL system.

The remainder of this paper is structured as follows: Section 2 outlines the research context and methodology, beginning with a review of TaRL implementation across regions such as India, Kenya, and Indonesia, followed by a synthesis of Morocco's national experience and its challenges. It then introduces the proposed DM-TaRL model and its four core design pillars: localized assessment, educator dashboards, real-time adaptivity, and gamification. Section 3 presents results from pilot data, platform comparisons, and a simulated use case. Section 4 discusses the findings in relation to global trends and implementation gaps. Section 5 concludes with key takeaways and recommendations for future deployment and research.

2. Materials and Methods

The Teaching at the Right Level (TaRL) approach has seen successful adoption in various countries, offering practical insights into both its challenges and outcomes, which are directly relevant to this study. TaRL was first launched in India in 2007 by the non-profit organization Pratham through its Read India program [6]. The method involved assessing students' actual learning levels and grouping them by ability rather than age or grade level. Teachers employed hands-on activities to build foundational skills in literacy and numeracy. This approach was integrated into the public education system through collaborations with state governments, resulting in measurable gains in student achievement.

The TaRL Africa initiative is managed by Empower Learning Africa, a not-for-profit organization registered and headquartered in Nairobi, Kenya, with teams in Côte d'Ivoire, Nigeria, and Zambia and locally registered offices in Nigeria and Côte d'Ivoire [7].

Beginning in 2016, Kenya implemented TaRL through partnerships with organizations like the Zizi Afrique Foundation. The focus was on enhancing core academic skills, particularly in rural areas. The application involved training teachers to assess students' competencies and group them based on learning levels. Regular assessments and tailored instructional activities were key components. The program led to noticeable gains in student performance and was subsequently scaled to reach more schools nationwide.

In sub-Saharan Africa, various countries have adopted the same approach over the past decade. The TaRL Nigeria program started in 2018, focusing on improving basic

literacy and numeracy skills. It was introduced in Ghana around 2014, primarily benefiting rural schools. The Uganda TaRL initiative began in 2015, focusing on addressing learning gaps in primary education.

In Southeast Asia, in 2021, Indonesia adapted the TaRL methodology into the All SMART Children (SAC) program, integrating it into public schools in Central Lombok with the support of INOVASI, an Australia–Indonesia partnership. The SAC approach retained TaRL’s core sequence, assessment, grouping, and instruction based on learning level while tailoring the implementation to local cultural and administrative contexts. In only two weeks, 35.4% of SAC participants advanced at least one level in reading, while the proportion of students in the lowest reading group dropped by over 6% [5]. The success of SAC, particularly its low-cost and scalable STMJ model (Systematic, Structured, Massive, and Affordable), underscores the adaptability of TaRL principles and informed the design of our Moroccan digital adaptation.

Morocco began piloting the TaRL approach during the 2022–2023 academic year [3]. An experimental study involving 106 pupils from 4th to 6th grade was conducted over a one-month remedial course, with daily 30 min sessions per subject. The results indicated statistically significant improvements in mathematics, Arabic, and French reading competencies, suggesting that TaRL effectively raised numeracy and literacy levels [8].

In 2024, a comprehensive remedial system based on TaRL principles was introduced in Moroccan schools to address fundamental knowledge gaps. Initial analyses of 63,000 pupils showed promising effects, with mastery rates multiplying by four in mathematics, by three in French, and by two in Arabic after eight weeks of remediation [9]. The implementation was further supported by the development of structured resources, including assessment and remediation exercises for each primary school grade.

However, the execution faced several contextual challenges that shaped the design of our digital solution. Teachers reported a steep learning curve when applying differentiated instruction, particularly in the absence of adequate support materials. Infrastructure deficits, especially in rural schools, limited the program’s scalability. The bilingual instructional setting (Arabic and French) required content adaptation, and a nationwide teacher boycott during post-testing disrupted standardized evaluation. These insights directly informed the design of our digital ecosystem—particularly its focus on real-time dashboards, automated diagnostics, and culturally responsive gamified content in both Arabic and French.

In response to these challenges and to support the sustainable scaling of TaRL, we propose DM-TaRL (Digitized Moroccan Teaching at the Right Level), a model that leverages modern educational technologies. Globally, digital tools have proven effective in personalizing instruction, supporting real-time assessments, and delivering interactive content. These capabilities have been shown to significantly improve student performance. These developments reinforce the potential of technology to complement and optimize the TaRL approach.

Our analysis identified four key pillars essential to effectively digitizing the TaRL system in Morocco:

- (1) Tailoring the assessment design to fit the Moroccan primary education context;
- (2) Developing an educator dashboard for monitoring and control;
- (3) Ensuring real-time adaptivity based on student levels and performance;
- (4) Incorporating gamification to sustain learner engagement.

These system requirements led us to conduct in-depth research on existing methods and solutions. Beginning with the first pillar, designing an effective assessment framework tailored to an education system requires alignment with curriculum standards while remaining culturally and contextually appropriate, as outlined by various studies [10].

Given Morocco's unique curriculum and linguistic diversity, where Arabic and French serve both as the primary languages of instruction, the proposed assessment system will default to Arabic, with French and English as secondary language options. Initial focus will be placed on core national curriculum subjects, such as Mathematics and Language Arts in Arabic and French, with plans to expand to additional disciplines including Biology, Geology, Physics, Chemistry, History, Geography, Civic Education, and Islamic Education.

To ensure comprehensive student evaluation, the system will incorporate both formative and summative assessment strategies. Formative assessments will provide continuous feedback through TaRL-based quizzes and exercises, fostering an adaptive learning experience. Meanwhile, summative assessments, such as mid-term and final exams, will remain under the purview of teachers and will not be digitized within the system.

Assessments must be inclusive and accessible, integrating visual elements, clear instructions, and well-defined evaluation criteria to accommodate diverse learning styles. Research on learning management systems (LMSs) [11] highlights the value of platforms such as Moodle, Google Classroom, and Microsoft Teams in enabling real-time tracking, collaborative learning, and data-driven instruction—elements that are central to a robust digital education system [12].

Digitizing an education system does not replace teachers; rather, it equips them with tools to improve efficiency. An educator dashboard enables real-time tracking of student performance, allowing teachers to adjust their strategies accordingly [13]. Studies show that teachers consult dashboards an average of 8.3 times per lesson to assess student progress and provide targeted feedback, promoting instructional responsiveness and data-driven decision-making [14]. Further research [15] suggests that teachers who balance human input with dashboard-generated insights deliver more effective feedback, reinforcing the importance of integrating technology into teaching practices.

Establishing an effective and user-friendly dashboard that can be easily accessed and utilized by educators, principals, and even parents presents several design challenges. Key issues include ensuring the interface is intuitive for all users and providing relevant data that can be easily interpreted at different levels. However, studies also highlight the opportunities such systems offer, particularly in fostering communication between stakeholders and supporting more personalized interventions.

A well-designed education system supports robust performance tracking and efficient resource management [16]. With built-in analytics, dashboards can generate timely insights that help educators fine-tune both teaching strategies and resource use [17,18].

While platforms like Khan Academy demonstrate the effectiveness of adaptive learning, where content difficulty adjusts dynamically based on student performance, more advanced systems such as Squirrel AI and DreamBox Learning take this further by continuously identifying student weaknesses and delivering targeted, real-time support, aligning with the adaptive sequencing proposed in our TaRL digitization framework. This personalization not only helps close knowledge gaps [19] but also allows students to progress at their own pace.

AI-powered personalization in dashboards has also been proven effective in improving outcomes without compromising learner engagement and retention, making it an invaluable tool even for mixed-reality and intelligent tutoring systems [20]. Platforms like Coursera and edX use AI-powered recommendation systems to customize learning pathways, expanding access to high-quality education globally.

Given that our system prioritizes the needs and experiences of both educators and students, we cannot overlook our fourth pillar: gamification. By incorporating game elements such as points, leaderboards, and badges, we aim to promote student engagement, motivation, and interactive learning [21]. Research has shown that gamification can transform

traditional classrooms into student-centric environments, fostering a sense of ownership over learning and encouraging collaboration [22].

Beyond motivation, gamification can significantly improve mathematical proficiency, particularly when combined with interactive problem-solving activities like coding [23]. To maximize its impact, gamified education systems must adapt continuously to meet student needs, ensuring inclusivity and sustained engagement [24].

Tools like Kahoot! and Classcraft have already demonstrated positive effects on student participation, though challenges in long-term engagement and curriculum alignment persist [25]. For successful implementation, however, it is essential to have both pedagogical and technological frameworks that support teachers in utilizing gamification strategies [26]. While students generally see value in gamified learning, teacher training is key to successful adoption [27]. Moreover, thoughtful design is necessary to avoid potential downsides such as excessive competition or disengagement among lower-performing students [28].

Gamifying our TaRL assessments will not only increase student engagement but also provide more accurate reflections of students' true abilities. By presenting the assessment as a game quest without explicitly identifying it as an evaluation, we can reduce anxiety and eliminate the potential for cheating.

This four-pillar framework, illustrated in Figure 1, forms the foundation of DM-TaRL, a strategic roadmap for Morocco's national digitized TaRL implementation. The model balances local constraints such as infrastructure, multilingual instruction, and teacher workload while maintaining a continued focus on assessing students based on their true learning levels.

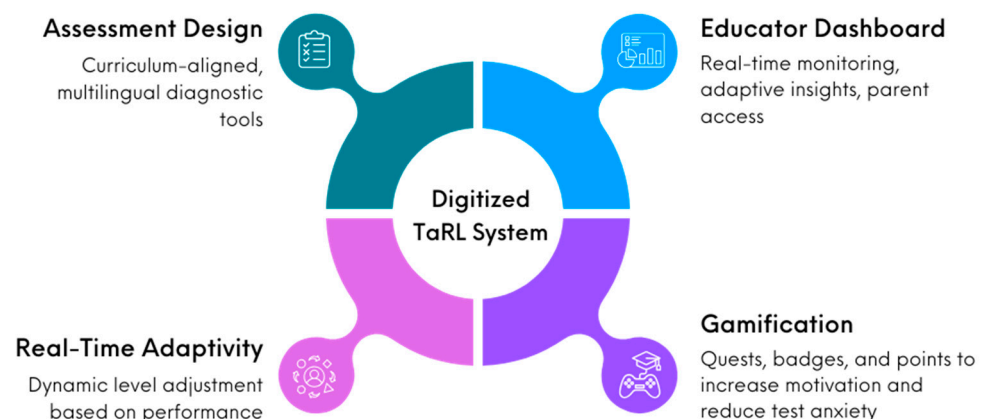


Figure 1. DM-TaRL: a four-pillar model for the implementation of Digitized Teaching at the Right Level in Morocco.

3. Results

As the Digitized Moroccan TaRL system is still under development, direct user-generated results are not yet available. However, a statistical analysis of remediation outcomes and a comparative review of educational platforms provide early insights into the potential impact of a digital TaRL system on Morocco's educational landscape. Additionally, a simulated case study projects how such a system could enhance student engagement, improve learning outcomes, and support teachers.

3.1. Statistical Evaluation of TaRL-Based Remediation Outcomes

To evaluate the impact of the Teaching at the Right Level (TaRL) approach in Moroccan primary education, we integrate national pilot study results [9] into our systematic framework. These findings, involving over 63,000 pupils and a verified subsample of

1400 students, illustrate marked progress in foundational competencies across Arabic, French, and Mathematics, as shown in Figure 2.

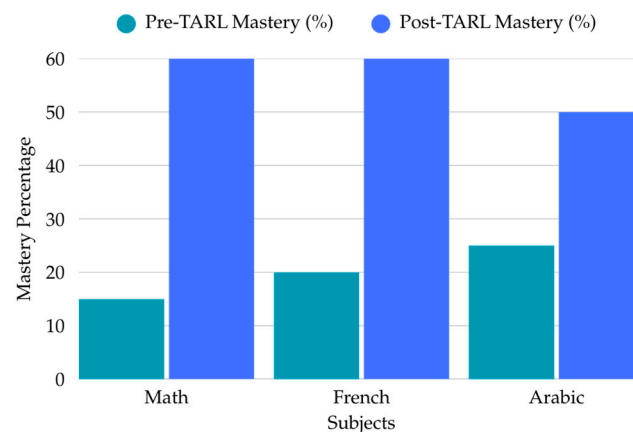


Figure 2. Mastery level gains by subject (pre- and post-TARL).

Mathematics shows the most significant improvement ($\times 4$), followed by French ($\times 3$) and Arabic ($\times 2$). These gains represent one to two years of academic catch-up within just two months of intervention, highlighting the capacity of TaRL to deliver accelerated foundational learning when properly implemented, as illustrated in Figure 3.

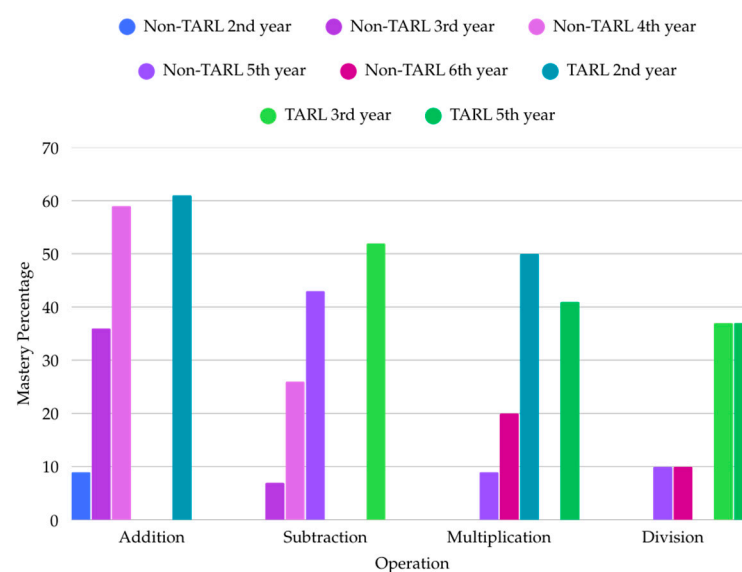


Figure 3. Comparison of skill mastery in math operations.

Students engaged in TaRL surpass their non-TARL peers by 1–2 grade levels in all fundamental math operations, confirming the method's effectiveness and scalability in improving numeracy outcomes.

Both Arabic and French literacy scores improved significantly. The mastery of fundamental decoding and comprehension abilities increased notably, positioning students to better navigate more complex curricular content, as demonstrated in Figure 4.

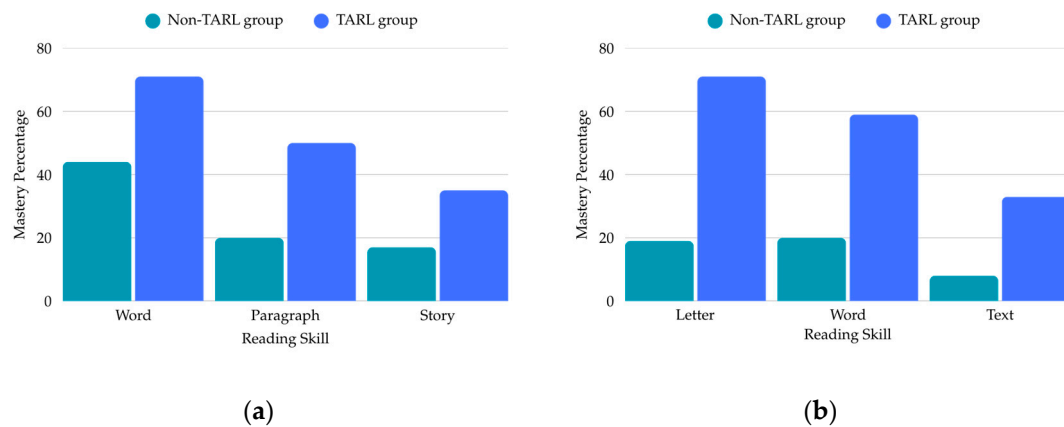


Figure 4. Literacy improvements in Arabic (a) and French (b).

3.2. Review and Comparative Analysis of Digital Tools Aligned with DM-TaRL Principles

3.2.1. Overview of Digital Tools Supporting DM-TaRL Approach

Several countries have experimented with digitized tools that align with the principles of the DM-TaRL model. These implementations integrate mobile learning apps, gamified content, and adaptive instruction to support literacy and numeracy development. Table 1 summarizes representative case studies from India, Kenya, and Indonesia, drawing from real-world deployment and NGO-led initiatives.

Table 1. Representative digital tools reflecting DM-TaRL pillars in India, Kenya, and Indonesia.

Country	Tools/Platforms	Modality	Outcomes	DM-TaRL Pillars Addressed
India	StoryWeaver, Read Along	Mobile digital library and speech-based reading app	Early literacy gains	Gamification, Real-Time Adaptivity
Kenya	Tusome Digital	Tablet-based assessments and teacher dashboards	Early literacy gains	Assessment Design, Educator Dashboard
Indonesia	Dawn of Civilization, Ed the Learning Bot	Gamified mobile learning app and AI-powered chatbot	Enhanced foundational skills	Gamification, Real-Time Adaptivity

In India, digital platforms such as StoryWeaver and Read Along have been instrumental in enhancing early literacy. StoryWeaver, developed by Pratham Books, is an open-access digital library offering thousands of multilingual children’s stories, enabling educators and learners to access culturally relevant reading materials [29]. Read Along, a Google-developed mobile application, utilizes speech recognition technology to assist children in reading aloud, providing real-time feedback and gamified incentives to encourage regular practice. Both platforms have contributed to improved reading engagement and foundational literacy skills among early-grade learners, in line with DM-TaRL’s principles emphasizing gamification, adaptive instruction, and culturally grounded content delivery.

In Kenya, the Tusome (Swahili for “Let’s Read”) program is a government-led initiative aimed at improving literacy outcomes in early primary grades. While it is not a direct digital implementation of the Teaching at the Right Level (TaRL) approach, it incorporates structured pedagogy, ongoing assessments, and tablet-supported dashboards—two pillars that align closely with the DM-TaRL framework: assessment design and educator dashboard. Research has shown that Tusome significantly improved reading fluency and comprehension among Grade 1 and 2 students through the use of teacher coaching, digital feedback systems, and national-level monitoring tools [30].

In the case of Indonesia, the Dawn of Civilization [31] and Ed the Learning Bot [32] platforms were implemented to support literacy and numeracy through gamified, adaptive learning in Indonesia [33]. While these tools were not developed as part of the official SAC program, they align closely with DM-TaRL's principles and were used in educational initiatives targeting similar foundational learning goals. Developed by Solve Education!, a Singapore-based nonprofit with regional offices and program deployments in Indonesia since 2017, these tools have documented impact through outreach reports and local education initiatives targeting out-of-school youth and underserved learners [34].

Each one of these tools shares core features with the DM-TaRL approach and has demonstrated measurable improvements among targeted learners. However, unlike the partial implementations seen in these case studies, DM-TaRL is founded on the full integration of all four pillars—localized assessment design, educator dashboard, real-time adaptivity, and gamification—to maximize its impact on learning outcomes.

3.2.2. Comparative Evaluation Against Popular Educational Platforms

To contextualize the DM-TaRL model within the broader landscape, we compare it against traditional TaRL implementations and established international digital learning systems. The comparison highlights key distinctions in adaptivity, cultural relevance, and teacher integration, as detailed in Table 2.

Table 2. Comparative analysis of traditional TaRL, global EdTech platforms, and the DM-TaRL model.

Platform/Method	Strengths	Limitations	Comparison with DM-TaRL
TaRL	Effective in grouping by learning level	High teacher workload; manual performance tracking	DM-TaRL automates diagnostics and reduces teacher burden
Squirrel AI	Strong personalization	Limited cultural relevance; minimal teacher involvement	DM-TaRL includes dashboards and localized content
Khan Academy	Excellent foundational content	Lacks real-time adaptivity	DM-TaRL adds dynamic level adjustment
DreamBox Learning	Adaptive and gamified	Weak teacher integration	DM-TaRL embeds teacher feedback and local alignment
Mindspark	Adaptive, multilingual, and evidence-based	Minimal gamification; requires supervised usage for impact	DM-TaRL adds gamification and real-time dashboards

While platforms such as Squirrel AI and DreamBox Learning offer advanced personalization and gamification, they often lack teacher involvement and contextual alignment with local curricula. Mindspark, despite being research-backed and adaptive, is most effective in structured, supervised settings. In contrast, DM-TaRL addresses these gaps by integrating adaptive, gamified learning technologies with educator dashboards and localized instructional design for scalable deployment in Moroccan classrooms.

Other platforms such as BYJU'S and ASSISTments offer subject-specific or visually rich content delivery, but their commercial models or narrow curricular focus limit their suitability for public, multilingual systems like Morocco's.

This comparative table underlines the need for a hybrid model that unites adaptability, localization, gamification, and real-time monitoring—features central to the digitized TaRL system proposed for Morocco.

3.3. Simulated Use Case

To illustrate the practical application of the ecosystem, the following case study in Table 3 simulates how a student interacts with the DM-TaRL system throughout the assessment and remediation cycles. The steps reflect the intended functionality of the system

under simulated conditions; actual performance may vary depending on contextual and technical implementation constraints.

Table 3. Simulated user journey in the DM-TaRL system.

Step	Process	System's Role	Expected Outcome
1. Student Profile	10-year-old student from rural Morocco with arithmetic difficulties	Uses initial profile to generate diagnostic test and learning plan	Early detection of gaps for personalized instruction
2. Diagnostic Assessment	Student completes a short game-based test	Places student in a skill-appropriate group	Accurate level placement and targeted support
3. Personalized Pathway	Student works through tailored activities	Adjusts tasks based on mastery and engagement	Confidence-building and gradual skill mastery
4. Gamified Progression	Student earns rewards for milestones	Provides badges, visuals, and game-like tracking	Motivation sustained through interactive feedback
5. Real-Time Teacher Monitoring	Teacher tracks progress live	Flags struggling areas and suggests instructional strategies	Timely interventions improve learning trajectory
6. Adaptive Learning Loop	Student continues through evolving tasks.	Reassesses periodically to refine path	Measurable progress in literacy and numeracy

4. Discussion

This work explored the feasibility and potential impact of a digitized Teaching at the Right Level framework tailored for Moroccan primary education. While field deployment is still pending, both the statistical evaluation of national pilot programs and the simulated user journey suggest that such a system represents a promising step toward addressing foundational learning deficits.

The key projected outcomes are notable: mastery levels in mathematics quadrupled, while French and Arabic showed significant gains following TaRL-based remediation.

Relative to international implementations, the Moroccan initiative distinguishes itself through its culturally adapted digital content and bilingual assessment framework. In India, for instance, mobile-based tools like StoryWeaver and Read Along focus on foundational literacy and personalization but lack diagnostic dashboards. Kenya's Tusome provided tablet-based assessments and educator dashboards but required continuous in-person support and coaching visits to ensure effective data usage, and this system does not incorporate gamification elements. In Indonesia, gamified tools like Dawn of Civilization and Ed the Learning Bot support foundational skills via mobile learning and AI-driven adaptivity, but they are not fully aligned with national curriculum standards and lack teacher-facing dashboards.

While tools like Squirrel AI and DreamBox Learning demonstrate effective personalization, they often lack cultural adaptability and educator integration. Unlike these platforms, the DM-TaRL system consolidates adaptive instruction, gamification, teacher analytics, and linguistic localization into a unified framework—bridging the gap between scalable technology and responsive pedagogy.

Despite these advantages, several limitations remain. DM-TaRL is still under development, and simulated data may not fully reflect the full complexity of real-world classroom environments. Furthermore, the international case studies reviewed vary in implementation fidelity, and their findings may not be entirely transferable to the Moroccan context. In addition, disparities in digital infrastructure, especially in rural schools, pose significant barriers to national scaling.

The findings nevertheless offer tangible insights. Teachers can benefit from reduced cognitive load and improved instructional targeting. Students gain access to interactive and

level-appropriate remediation that adapts to their progress. Policymakers may leverage real-time learning analytics to support data-driven decision-making at scale. These affordances collectively support Morocco's national goals of educational equity and quality.

Future research should prioritize pilot deployments to assess real-world usability, engagement patterns, and learning outcomes. It is also recommended to explore longitudinal effects, teacher adoption dynamics, and the policy mechanisms needed to support systemwide scaling. Special attention must be paid to ensuring inclusivity for learners with limited access to devices or connectivity.

By combining TaRL's proven pedagogical value with modern educational technology, this study presents a pathway for scalable and context-aware innovation in foundational learning. The Digitized Moroccan TaRL initiative may offer a blueprint for other Global South countries pursuing equitable and sustainable educational transformation.

5. Conclusions

This study contributes to ongoing efforts to improve foundational learning through the digitization of the Teaching at the Right Level (TaRL) approach in Moroccan primary schools. By aligning assessment, instruction, and real-time monitoring within a unified digital system, the proposed model offers a practical response to persistent gaps in early education.

The literature reviewed highlights key global trends, including the rise of gamified, adaptive, and teacher-informed digital tools. However, most existing systems lack full integration of all four pillars defined by DM-TaRL: localized assessment, educator dashboards, real-time adaptivity, and gamification. In contrast, the Moroccan initiative combines these elements within a bilingual, culturally tailored framework designed for scalable implementation.

While the system remains under development, the integration of these features offers a solid foundation for future deployment. As Morocco advances its efforts in equitable education reform, this model may serve as a guiding structure for enhancing engagement and learning outcomes in public classrooms.

Future work will focus on testing a functional DM-TaRL prototype with students and teachers in rural Moroccan schools, leveraging the support and approval already secured from a regional educational inspector. This phase will help evaluate real-world usability, teacher engagement, and student progress. Further research will prioritize continuous system development and refinement, including dynamic content generation and evaluation integration through large language models. These advances will be essential to ensuring long-term relevance, personalization, and pedagogical impact in diverse learning contexts.

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Abbreviations

The following abbreviations are used in this manuscript:

DM-TaRL	Digitized Moroccan Teaching at the Right Level
TaRL	Teaching at the Right Level
LMS	Learning management system
AI	Artificial Intelligence
NGO	Non-Governmental Organization
CNRST	Centre National pour la Recherche Scientifique et Technique
FSTT	Faculty of Sciences and Technologies of Tangier
C3S	Computer Science and Smart Systems

References

1. Yabiladi.com. Post-COVID-19: The World Bank Calls for a «More Solid» Moroccan Educational System. Available online: <https://en.yabiladi.com/articles/details/101046/post-covid-19-world-bank-calls-more.html> (accessed on 30 March 2025).
2. UNESCO. Addressing Students' Learning Gaps to Ensure Quality Learning for All Children and Prevent Drop-Out in Asia-Pacific. Available online: <https://www.unesco.org/en/articles/addressing-students-learning-gaps-ensure-quality-learning-all-children-and-prevent-drop-out-asia> (accessed on 16 April 2025).
3. Masaiti, A.E. Morocco to Adopt India's Teaching at the Right Level Educational System. HESPRESS English—Morocco News. Available online: <https://en.hespress.com/50958-morocco-to-adopt-indias-teaching-at-the-right-level-educational-system.html> (accessed on 30 March 2025).
4. Ismail, I.A.; Qadhafi, Jhora, F.U.; Yorinda, A. Teaching at the Right Level (TaRL) as a Potential Solution for Improving Middle School Education: A Systematic Review of the Literature. *Int. J. Acad. Pedagog. Res.* **2024**, *8*, 126–138. [CrossRef]
5. Jazuli, L. Teaching at the Right Level (TaRL) Through the All Smart Children Approach (SAC) Improves Student'S Literature Ability. *Prog. Pendidik.* **2024**, *3*, 156–165. [CrossRef]
6. Pratham. Teaching at the Right Level. Available online: <https://www.pratham.org/about/teaching-at-the-right-level/> (accessed on 16 April 2025).
7. TaRL Africa Teaching at the Right Level—Africa. Available online: <https://teachingattherightlevel.org/> (accessed on 16 April 2025).
8. Binaoui, A.; Moubtassime, M.; Belfakir, L. The Effectiveness of the TaRL Approach on Moroccan Pupils' Mathematics, Arabic, and French Reading Competencies. *Int. J. Educ. Manag. Eng.* **2023**, *13*, 1–10. [CrossRef]
9. Akdi, O.; Belamhitou, M.M. The Teaching at the Right Level approach: A paradigm shift to accelerate Moroccan pupil's learning. *Afr. Educ. Res. J.* **2024**, *12*, 238–254.
10. Chen, C.-H.; Tzeng, G.-H. Creating the aspired intelligent assessment systems for teaching materials. *Expert Syst. Appl.* **2011**, *38*, 12168–12179. [CrossRef]
11. Kruger, D.; Inman, S.; Ding, Z.; Kang, Y.; Kuna, P.; Liu, Y.; Lu, X.; Oro, S.; Wang, Y. Improving Teacher Effectiveness: Designing Better Assessment Tools in Learning Management Systems. *Future Internet* **2015**, *7*, 484–499. [CrossRef]
12. Edson, A.J.; Phillips, E.D. Connecting a teacher dashboard to a student digital collaborative environment: Supporting teacher enactment of problem-based mathematics curriculum. *ZDM Math. Educ.* **2021**, *53*, 1285–1298. [CrossRef]
13. Molenaar, I.; Knoop-van Campen, C. Teacher Dashboards in Practice: Usage and Impact. In *Proceedings of the Data Driven Approaches in Digital Education*; Lavoué, É., Drachsler, H., Verbert, K., Broisin, J., Pérez-Sanagustín, M., Eds.; Springer International Publishing: Cham, Switzerland, 2017; pp. 125–138. [CrossRef]
14. Lee-Cultura, S.; Sharma, K.; Giannakos, M.N. Multimodal Teacher Dashboards: Challenges and Opportunities of Enhancing Teacher Insights Through a Case Study. *IEEE Trans. Learn. Technol.* **2024**, *17*, 181–201. [CrossRef]
15. Knoop-van Campen, C.; Molenaar, I. How Teachers Integrate Dashboards into Their Feedback Practices. *Frontline Learn. Res.* **2020**, *8*, 37–51. [CrossRef]
16. Williamson, B. Digital education governance: Data visualization, predictive analytics, and 'real-time' policy instruments. *J. Educ. Policy* **2016**, *31*, 123–141. Available online: <https://www.tandfonline.com/doi/abs/10.1080/02680939.2015.1035758> (accessed on 21 April 2025). [CrossRef]

17. Moltudal, S.H.; Krumsvik, R.J.; Høydal, K.L. Adaptive Learning Technology in Primary Education: Implications for Professional Teacher Knowledge and Classroom Management. *Front. Educ.* **2022**, *7*, 830536. [CrossRef]
18. Raj, N.S.; Renumol, V.G. An improved adaptive learning path recommendation model driven by real-time learning analytics. *J. Comput. Educ.* **2024**, *11*, 121–148. [CrossRef]
19. Debeer, D.; Vanbecelaere, S.; Noortgate, W.V.D.; Reynvoet, B.; Depaepe, F. The effect of adaptivity in digital learning technologies. Modelling learning efficiency using data from an educational game. *Br. J. Educ. Technol.* **2021**, *52*, 1881–1897. [CrossRef]
20. Yannier, N.; Hudson, S.E.; Chang, H.; Koedinger, K.R. AI Adaptivity in a Mixed-Reality System Improves Learning. *Int. J. Artif. Intell. Educ.* **2024**, *34*, 1541–1558. [CrossRef]
21. Cunha, G.C.A.; Barraqui, L.P.; de Freitas, S.A.A. Evaluating the use of gamification in mathematics learning in primary school children. In Proceedings of the 2018 IEEE Frontiers in Education Conference (FIE), San Jose, CA, USA, 3–6 October 2018; pp. 1–4. [CrossRef]
22. Halloluwa, T.; Vyas, D.; Usoof, H.; Hewagamage, K.P. Gamification for development: A case of collaborative learning in Sri Lankan primary schools. *Pers. Ubiquit. Comput.* **2018**, *22*, 391–407. [CrossRef]
23. Folgieri, R.; Vanutelli, M.E.; Galbiati, P.D.V.; Lucchiari, C. Gamification and Coding to Engage Primary School Students in Learning Mathematics: A Case Study. In Proceedings of the 11th International Conference on Computer Supported Education, Heraklion, Crete, Greece, 2–4 May 2019; SciTePress: Setúbal, Portugal, 2019. [CrossRef]
24. Arufe-Giráldez, V.; Sanmiguel-Rodríguez, A.; Ramos-Álvarez, O.; Navarro-Patón, R. Gamification in Physical Education: A Systematic Review. *Educ. Sci.* **2022**, *12*, 540. [CrossRef]
25. Dimitra, K.; Konstantinos, K.; Christina, Z.; Katerina, T. Types of Game-Based Learning in Education: A Brief State of the Art and the Implementation in Greece. *Eur. Educ. Res.* **2020**, *3*, 87–100.
26. Vrcelj, A.; Hoic-Božic, N.; Dlab, M.H. Use of Gamification in Primary and Secondary Education: A Systematic Literature Review. *Int. J. Educ. Methodol.* **2023**, *9*, 13–27. [CrossRef]
27. Fiuza-Fernández, A.; Lomba-Portela, L.; Soto-Carballo, J.; Pino-Juste, M.R. Study of the knowledge about gamification of degree in primary education students. *PLoS ONE* **2022**, *17*, e0263107. [CrossRef] [PubMed]
28. Participatory Educational Research» Submission» The Views and Adoption Levels of Primary School Teachers on Gamification, Problems and Possible Solutions. Available online: <https://dergipark.org.tr/en/pub/per/issue/54375/713777> (accessed on 19 March 2025).
29. Pratham Books. StoryWeaver. StoryWeaver Has Been Recognised as a Digital Public Good for Foundational Literacy and Early-Grade Reading! Available online: <https://storyweaver.org.in> (accessed on 19 March 2025).
30. Piper, B.; Destefano, J.; Kinyanjui, E.M.; Ong’ele, S. Scaling up successfully: Lessons from Kenya’s Tusome national literacy program. *J. Educ. Chang.* **2018**, *19*, 293–321. [CrossRef]
31. Solve Education! Dawn of Civilization—Play and Learn English! Dawn of Civilization. Available online: <https://dawnofcivilization.net/> (accessed on 16 April 2025).
32. Solve Education! Technology—ED THE LEARNING BOT. Available online: <https://solveeducation.org/technology/ed-the-learning-bot/> (accessed on 16 April 2025).
33. Fuadah, N.T. Solve Education! Supporting Indonesia’s Educational Transformation at GSVI 2024. Solve Education! Available online: <https://solveeducation.org/solve-education-supporting-indonesias-educational-transformation-at-gsvi-2024/> (accessed on 16 April 2025).
34. Solve Education! Solve Education! Annual Progress Report 2022. 2022. Available online: https://solveeducation.org/wp-content/uploads/Annual_Progress_Report_2022.pdf (accessed on 19 March 2025).

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